

The L^AT_EX Tipbook

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1 Common mistakes

L^AT_EX for the beeeeeeeest, yeah,


Felix Halbwedl

Quotation marks. T_EX does not automatically recognise left and right quotation marks. By convention, a left single quotation mark is written with ``` (on a British keyboard, it is the key to the left of the number 1) while a right single quotation mark is `'`. For double quotation marks, the convention is to repeat the corresponding single quotation marks twice. Knowledge of this automatically grants one bragging rights as even T_EX experts use the wrong quotation marks all the time.

T_EX and L^AT_EX. When typing T_EX and L^AT_EX, one should always wrap them in curly brackets. Otherwise things will go wrong in your T_EXdocument¹.

2 Tricks

I have used many questionable packages. Questionable in the sense that they do not work as they should.

Ketevan Arabuli

Zero indentation. Write

```
\parindent=0pt
```

in the preamble.

Multiple packages in one line. Write

```
\usepackage{package1,package2,package3}
```

in the preamble.

Normal ordering. There is no in-built command for normal ordering. To implement it, write

```
\newcommand{\normalordering}[1]{\mathopen{:}\mathrel{\#1}\mathclose{:}}
```

or something similar in the preamble.

¹See?

European date format. By default, L^AT_EX uses the North American date format. For the European date format, write

```
\usepackage[UKenglish]{isodate}
\cleanlookdateon % optional
```

or something similar in the preamble.

Verbatim. In environments like `\footnote` the default `verbatim` does not work. For a solution, use the package `fancyvrb`. For full customisation, use `listings`.

Big brackets. Sometimes `\left` and `\right` are impossible due to line breaking, in which case, simply use commands like `\biggl` and `\biggr`.

Big horizontal bracket. Write

```
\underbrace{some expression}_{some annotation}
```

Scaling text arbitrarily. This trick is twofold:

- To scale your text to an arbitrary font size, write `\fontsize{one}{two}your text` where, by convention, `two` should be 1.2 times the size of `one`.
- To scale your text w.r.t. your `textwidth`, write `\resizebox{percentage\textwidth}{!}{your text}` where `!` auto-scales the vertical component.

Einstein summation convention. In some conventions, the order of indices must be made apparent, like $R_{\nu\lambda}^{\mu\sigma}$. To do so, write

```
R^{\mu}_{\nu}_{\lambda}^{\sigma}
```

Text superscript. To implement superscripts in text mode, like 2nd, write

```
2\textsuperscript{nd}
```

3 Eye candy

L^AT_EX takes care of the prettiness automatically, so you have more time to focus on the content. Which is the thing that in the end matters most.

Monika Schied

Circled text. Use the package `circledsteps`.

To-do notes. Use the package `todonotes`.

Customised titles. Use the package `titlesec`.

Vector graphics. .SVG files on their own do not work in most cases. A more realistic approach is to convert .SVG files into .PDF files, which can then be inserted as-is.

Colour boxes. A quick solution is `mdframed`. This package suffices for most use cases, but can be buggy when theorems are nested into them. In addition, this package disallows true transparency, which is problematic if the document has watermarks. For a more sophisticated solution, use `tcolorbox` in conjunction with packages like `amsthm`.

Smol math spaces. Within the math environment *smol* spaces are achieved via `\,, \;`, etc.

Extra vertical space. Write

```
\vspace{height of vertical space}
```

Vertical centring. Write

```
\vspace*{\fill}
% your stuff here
\vspace*{\fill}
```

Scaled pictures in text. This can be accomplished by defining new commands:

```
\usepackage{scalerel}
\NewDocumentCommand\yourcommand{}{\scalerel*{\includegraphics{yourpic}}{X}}
```

Credit: Monika Schied.

Picture as whole page. Write

```
\begin{tikzpicture}[remember picture,overlay]
\node[inner sep=0pt] (cover) at (current page.center){\includegraphics[width=\paperwidth]{yourimage}}
\end{tikzpicture}
```

Accents. Useful for names that are ‘foreign’ w.r.t. the keyboard format of your country. If one is too lazy to search for special characters, $\text{\LaTeX}$ has a way of rendering them. e.g.:

- Christian G. B\''ohmer $\rightarrow$ Christian G. Böhmer
- Emilia Szyma\'nska $\rightarrow$ Emilia Szymańska
- Hermann Gra{\ss}mann $\rightarrow$ Hermann Graßmann
- Georges Lema\^itre $\rightarrow$ Georges Lemaître
- Hans Christian {\O}rsted $\rightarrow$ Hans Christian Ørsted

and so on.

4 Conferences

Stuff like posters and presentations are shit in $\text{\LaTeX}$ TBH. As someone who has tried designing a poster in $\text{\LaTeX}$ I am allowed to say that.

Thomas Hutter, who organises the so-called $\text{\LaTeX}$ -Tutorien

Posters. Use the class `tikzposter`.

Beamer. For a full manual of Beamer basics, see [here](#). Notably, one should recognise that they may be well capable of producing a custom Beamer style that far surpasses the official one of their institution - assuming said institution has a style to begin with.

Bottom strip. To show alternate titles and authors on the strip that appears on the bottom of Beamer slides in some styles, write

```
\title[Alternate]{Title Page}  
\author[Alternate]{Title Page}
```

5 Trolling

The sword of T_EX yields the glory
of the physicist!

First author

Aphorisms. Use the package `epigraph`.

Emoticons. Use the package `emoji`.

Animals. Use the package `tikzlings` and `tikzducks`.

Figurines. Use the package `tikzpeople`.

Comic Sans. The `comicsans` package is out of date and must be used simultaneously with `tccomicsans` (link [here](#)). Unfortunately, the latter is not standard issue on Overleaf due to it not being on CTAN, and must be installed manually. One more reason to use a local T_EX compiler instead of Overleaf.

6 Good stuff

A bit of history. T_EX was preceded by [groff](#). Before the emergence of .PDF files as the de facto standard, the common T_EX output format was .DVI.

Senan. The T_EX mentor of the first author's T_EX mentor. Even though he ultimately joined the financial sector, his magnum opus can still be seen [here](#).

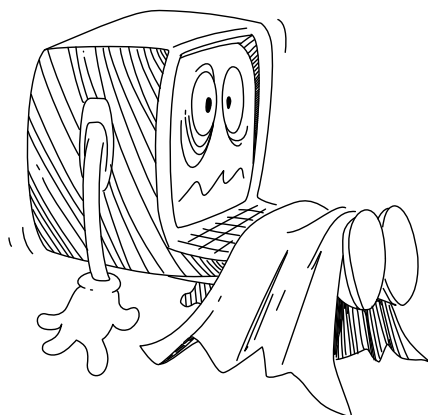


Figure 1: Average T_EX user.